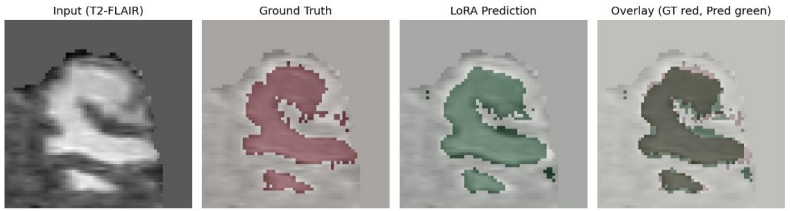
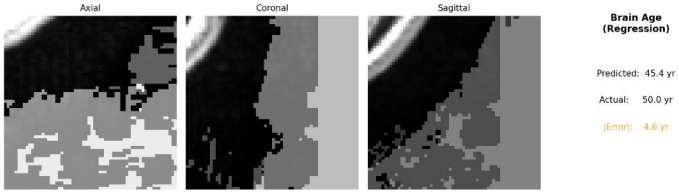


Figure 1: Framework: frozen pretrained backbone f_θ with task-specific LoRA adapters ϕ_k and heads h_k. Gray = frozen; blue = trainable per task. At inference for T1 (resp. T2): backbone + ϕ_1 + h_1 (resp. ϕ_2 + h_2).

Task1: Tumor Segmentation



Task2: Brain Age Estimation



BWT=0

Zero Catastrophic Forgetting

Frozen backbone + isolated LoRA adapters guarantee no forgetting by design

< 0.1%

Parameter Efficient Only

~28K-47K trainable params per task (vs. 59M backbone) ~1.7 GB peak GPU memory

n = 32

Few-Shot Ready Competitive

performance with as few as 32 labeled samples per downstream task